

Lab - .NET - Blob Metadata

To Begin with the Lab

Summary

In this lab, you learn how to **set and retrieve metadata** for a blob stored in Azure Blob Storage using a .NET application. Metadata is stored as **key-value pairs** and provides additional information about a blob without changing its contents. The application uses `SetMetadataAsync()` to add metadata and `GetPropertiesAsync()` to retrieve and display it.

Understanding

Blob Metadata is user-defined information associated with a blob. Unlike the file content, metadata is not part of the actual file; instead, it describes the file by storing details such as the owner, course name, project, or version.

To work with metadata, the application first creates a **BlobClient** for the required blob. The `SetMetadataAsync()` method stores metadata in Azure Blob Storage, replacing any existing metadata on the blob. The `GetPropertiesAsync()` method retrieves the blob's properties, including the metadata. The metadata is then displayed by looping through the **Metadata** collection.

Metadata is commonly used for categorizing files, organizing application data, and filtering blobs based on custom attributes.

Example

Suppose you have a blob named **01.py** stored inside the **images** container.

Blob:

01.py

Metadata:

owner = cloudeus

course = az-204

lab = blob-metadata

Application Flow:

.NET Application



BlobClient



Blob (01.py)



Metadata

├── owner = cloudeus

├── course = az-204

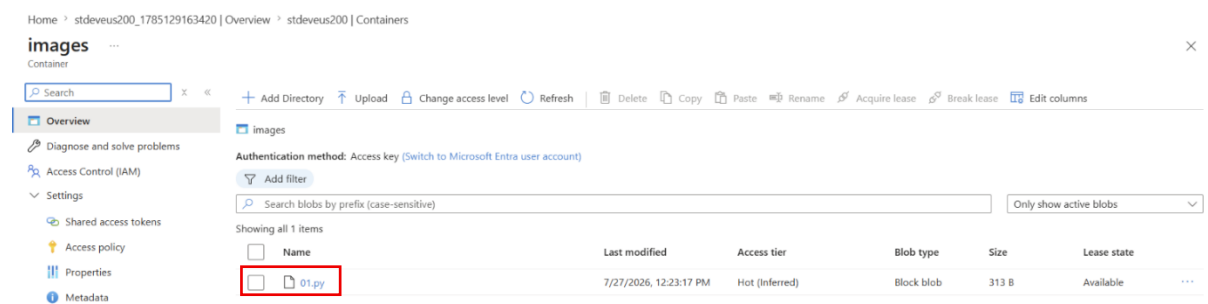
└── lab = blob-metadata

When the application runs, it stores the metadata with the blob and later retrieves and displays each metadata key and value.

Lab Steps

Step 1: Open the .NET Project

- Open your .NET project in Visual Studio or Visual Studio Code.
- Ensure the Azure Blob Storage SDK is installed.
- Verify that the blob **01.py** already exists in the Blob Container.



Step 2: Create the SetMetaData Method

Create a method to add metadata to the blob.

```
static async Task SetMetaData(BlobContainerClient containerClient)
{
    string blobName = "01.py";

    var blobClient = containerClient.GetBlobClient(blobName);

    var metadata = new Dictionary<string, string>
    {
        ["owner"] = "cloudxeus",
        ["course"] = "az-204",
        ["lab"] = "blob-metadata"
    };

    await blobClient.SetMetadataAsync(metadata);

    Console.WriteLine("Metadata set");
}
```

- Create a BlobClient.
- Create a metadata dictionary.
- Add metadata as key-value pairs.
- Store the metadata using SetMetadataAsync().

```
powershell blobapp Program.cs X
blobapp > Program.cs
44 static async Task DownloadBlob(BlobContainerClient containerClient)
56
57     Console.WriteLine("Blob downloaded successfully");
58 }
59
60 //await DownloadBlob(containerClient);
61
62 static async Task SetMetadata(BlobContainerClient containerClient)
63 {
64     string blobName="01.py";
65     var blobClient= containerClient.GetBlobClient(blobName);
66     var metadata = new Dictionary<string, string>
67     {
68         ["owner"] = "cloudxeus",
69         ["course"] = "az-204",
70         ["lab"] = "blob-metadata"
71     };
72
73     await blobClient.SetMetadataAsync(metadata);
74     Console.WriteLine("Metadata set");
75
76 }
```

Step 3: Call the SetMetadata Method (Optional)

Call the method to save metadata to the blob.

```
await SetMetadata(containerClient);
```

- Connects to the blob.
- Saves the metadata.
- Displays **Metadata set** after successful execution.

Note: In the provided code, this method is commented out. Uncomment it when you want to update or add metadata.

```
static async Task SetMetadata(BlobContainerClient containerClient)
{
    string blobName="01.py";
    var blobClient= containerClient.GetBlobClient(blobName);
    var metadata = new Dictionary<string, string>
    {
        ["owner"] = "cloudxeus",
        ["course"] = "az-204",
        ["lab"] = "blob-metadata"
    };

    await blobClient.SetMetadataAsync(metadata);
    Console.WriteLine("Metadata set");
}

await SetMetadata(containerClient);
```

Step 4: Create the GetMetadata Method

Create a method to retrieve metadata from the blob.

```
static async Task GetMetadata(BlobContainerClient containerClient)
{
    string blobName = "01.py";

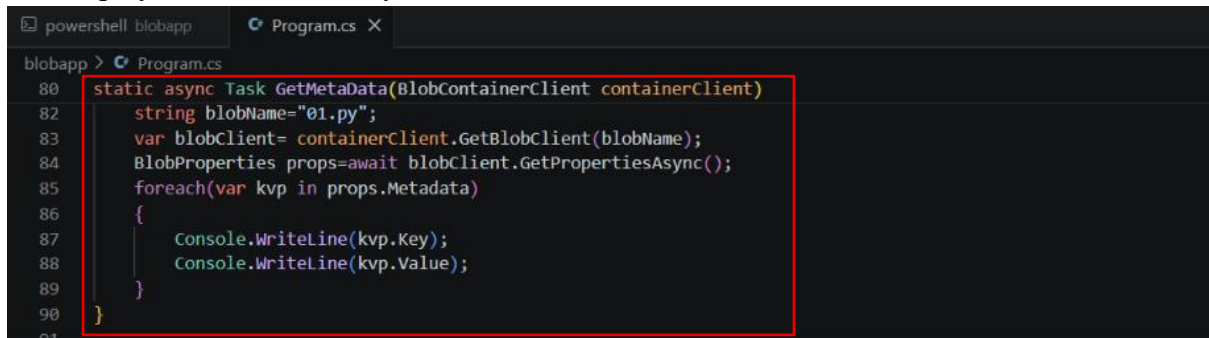
    var blobClient = containerClient.GetBlobClient(blobName);

    BlobProperties props = await blobClient.GetPropertiesAsync();

    foreach (var kvp in props.Metadata)
    {
        Console.WriteLine(kvp.Key);
    }
}
```

```
        Console.WriteLine(kvp.Value);
    }
}
```

- Create a BlobClient.
- Retrieve blob properties.
- Access the metadata collection.
- Display each metadata key and value.



```
powershell blobapp  Program.cs X
blobapp > Program.cs
80 static async Task GetMetadata(BlobContainerClient containerClient)
82     string blobName="01.py";
83     var blobClient= containerClient.GetBlobClient(blobName);
84     BlobProperties props=await blobClient.GetPropertiesAsync();
85     foreach(var kvp in props.Metadata)
86     {
87         Console.WriteLine(kvp.Key);
88         Console.WriteLine(kvp.Value);
89     }
90 }
```

Step 5: Call the GetMetadata Method

Retrieve and display the blob metadata.

```
await GetMetadata(containerClient);
```

- Reads the metadata stored on the blob.
- Displays all metadata keys and values in the console.



```
powershell blobapp  Program.cs X
blobapp > Program.cs
80 static async Task GetMetadata(BlobContainerClient containerClient)
82     string blobName="01.py";
83     var blobClient= containerClient.GetBlobClient(blobName);
84     BlobProperties props=await blobClient.GetPropertiesAsync();
85     foreach(var kvp in props.Metadata)
86     {
87         Console.WriteLine(kvp.Key);
88         Console.WriteLine(kvp.Value);
89     }
90 }
91
92 await GetMetadata(containerClient);
```

Step 6: Build the Project

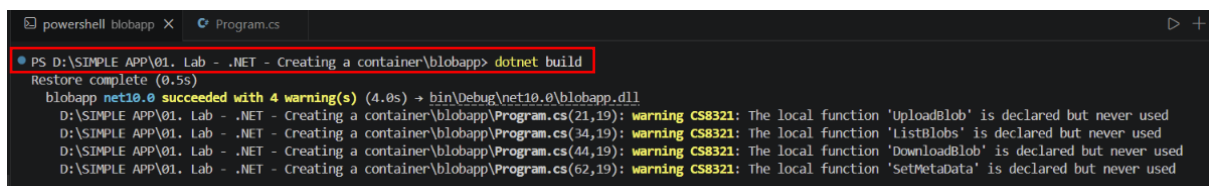
Run the following command:

```
dotnet build
```

- Compiles the application.
- Verifies that there are no build errors.

Expected Output:

Build succeeded.



```
powershell blobapp  Program.cs X
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> dotnet build
Restore complete (0.5s)
blobapp net10.0 succeeded with 4 warning(s) (4.0s) -> bin\Debug\net10.0/blobapp.dll
D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> Program.cs(21,19): warning CS8321: The local function 'UploadBlob' is declared but never used
D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> Program.cs(34,19): warning CS8321: The local function 'ListBlobs' is declared but never used
D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> Program.cs(44,19): warning CS8321: The local function 'DownloadBlob' is declared but never used
D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> Program.cs(62,19): warning CS8321: The local function 'SetMetadata' is declared but never used
```

Step 7: Run the Application

Run the following command:

dotnet run

- Starts the .NET application.
- Stores metadata (if SetMetaData() is called).
- Retrieves and displays the metadata.

Example Output:

Metadata set

owner

cloudxeus

course

az-204

lab

blob-metadata

```
powershell blobapp X Program.cs
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> dotnet build
Restore complete (0.5s)
blobapp net10.0 succeeded with 3 warning(s) (4.0s) -> bin\Debug\net10.0/blobapp.dll
D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp\Program.cs(21,19): warning CS8321: The local function 'UploadBlob' is declared but never used
D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp\Program.cs(34,19): warning CS8321: The local function 'ListBlobs' is declared but never used
D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp\Program.cs(44,19): warning CS8321: The local function 'DownloadBlob' is declared but never used
Build succeeded with 3 warning(s) in 5.1s
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> dotnet run
Container created
Metadata set
owner
cloudxeus
course
az-204
lab
blob-metadata
PS D:\SIMPLE APP\01. Lab - .NET - Creating a container/blobapp> |
```

Step 8: Verify Metadata in Azure Portal

- Open the Azure Portal.
- Navigate to your **Storage Account**.
- Select **Data Storage** → **Containers**.
- Open the **images** container.
- Select the blob **01.py**.
- Open the **Metadata** tab.
- Verify that the metadata has been added successfully.

The screenshot shows the Azure Portal interface. On the left, the 'images' container is selected. In the center, the '01.py' blob is selected. On the right, the 'Metadata' tab is open, displaying a table of metadata.

Key	Value
owner	cloudxeus
course	az-204
lab	blob-metadata